

Lucas Nunes

POSTDOCTORAL RESEARCHER · RWTH AACHEN UNIVERSITY

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Research Interests

Generative models · Representation learning · 3D scene understanding · Perception

Education

Dr.-Ing. (Ph.D.) Degree - *summa cum laude* / with distinction

University of Bonn

THESIS: "LEARNING DISCRIMINATIVE REPRESENTATIONS AND GENERATIVE APPROACHES FOR OUTDOOR 3D LIDAR DATA"

Feb/2026

(SUPERVISOR: PROF. DR. CYRILL STACHNISS)

M.Sc. in Computer Science

University of São Paulo

THESIS: "ENVIRONMENT RECONSTRUCTION ON DISPARITY IMAGES USING SURFACE FEATURES AND GENERATIVE ADVERSARIAL NETWORKS"

Mar/2020

(SUPERVISOR: PROF. DR. DENIS F. WOLF)

Bachelor in Computer Science

University of São Paulo

DEGREE IN COMPUTER SCIENCE

Jan/2018

Academic Positions

Postdoctoral Researcher

RWTH Aachen University

COMPUTER VISION GROUP

Mar/2026 - Present

Ph.D. Student and Research Assistant

University of Bonn

PHOTOGRAMMETRY & ROBOTICS LAB, INSTITUTE OF GEODESY AND GEOINFORMATION

Nov/2020 - Dec/2025

Intern Researcher

Karlsruhe Institute of Technology

INSTITUT FÜR MESS- UND REGELUNGSTECHNIK

Aug/2018 - Nov/2018

Master Student and Research Assistant

University of São Paulo

MOBILE ROBOTICS LABORATORY, INSTITUTE OF MATHEMATICS AND COMPUTER SCIENCE

Feb/2018 - Mar/2020

Junior Research Assistant

University of São Paulo

MOBILE ROBOTICS LABORATORY, INSTITUTE OF MATHEMATICS AND COMPUTER SCIENCE

Ago/2014 - Jun/2017

Awards

Faculty Award 2026 for Geodesy for *Towards Generating Realistic 3D Semantic Training Data for Autonomous Driving* (L. Nunes et al., PAMI)

2026

Finalist - IROS 2025 Best Agri-Robotics Paper Award for *Zero-Shot Semantic Segmentation for Robots in Agriculture*

2025

Teaching

Mobile Sensing and Robotics I - University of Bonn, MSc, 2h/week, shared teaching, winter term

2025

Mobile Sensing and Robotics II - University of Bonn, MSc, 2h/week, lecture, summer term

2025

Mobile Sensing and Robotics II - University of Bonn, MSc, 2h/week, shared teaching, summer term

2024

Techniques for Self-Driving Cars - University of Bonn, MSc, 2h/week, single lecture, winter term

2023

Open-World Panoptic Segmentation of Traffic Participants - University of Bonn, MSc, project, every term

2023

Machine Learning for Robotics and Computer Vision - University of Bonn, MSc, 2h/week, tutoring, summer term

2022

Machine Learning for Robotics and Computer Vision - University of Bonn, MSc, 2h/week, tutoring, summer term

2021

Online Teaching Examples (Links to Youtube Videos)

Self-Driving Cars: Unsupervised Learning for Vehicles Perception

Link: <https://www.youtube.com/watch?v=9KA04ayP2P4>

2023

Supervision

Novel View Synthesis of Indoor Dynamic Scenes

University of Zurich, MSc thesis

2024

Fruits 3D Reconstruction for Agricultural Robotics with Diffusion Models

University of Bonn, MSc thesis

2025

Invited Talks

Data Science Sanca Meetup. São Carlos, Brazil on *How Autonomous Vehicles can perceive what is happening around and make decisions?*

Mar/2018

4CORP Meetup, Futurecom. São Paulo, Brazil. on *The Autonomous Vehicles Revolution: A New Relation Between User and Vehicle*

Out/2019

Academic Services

REVIEWING SERVICES

Reviewer for conferences: IROS, ICRA, IV, ICCV, NeurIPS, CVPR, ECCV, RSS

Reviewer for Journals: IEEE Robotics and Automation Letters, IEEE Transactions on Robotics, IEEE Transactions on Pattern Analysis and Machine Intelligence

WORKSHOP ORGANIZATION

ICRA Workshop on *Uncertainty in Open-World Robotics*

2026

Professional Experience

Software Engineer

DEVELOPMENT OF FRONT AND BACK-END FOR SAFETY ASSESSMENT TOOL AND URBAN SIMULATED ENVIRONMENTS IN THE CONTEXT OF AUTONOMOUS VEHICLES

IVEX
Apr/2020 – Nov/2020

Technical Leader

COORDINATION OF A WEB CRAWLER PROGRAMMING TEAM FOR DATA COLLECTION AND DEVELOPMENT/MAINTENANCE OF A BACK-END DATABASE SERVER

Juristec+
Jan/2019 – Apr/2020

Python Developer

DEVELOPMENT OF WEB CRAWLER PYTHON SCRIPTS FOR DATA COLLECTION

Juristec+
Jan/2018 – Aug/2018

Publication List

Peer-reviewed journal articles

- [1] L. Nunes, R. Marcuzzi, J. Behley, and C. Stachniss, "Towards Generating Realistic 3D Semantic Training Data for Autonomous Driving," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026.
- [2] R. Marcuzzi, L. Nunes, E. Marks, L. Wiesmann, T. Läbe, J. Behley, and C. Stachniss, "SfmOcc: Vision-Based 3D Semantic Occupancy Prediction in Urban Environments," *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [3] L. Wiesmann, T. Läbe, L. Nunes, J. Behley, and C. Stachniss, "Joint Intrinsic and Extrinsic Calibration of Perception Systems Utilizing a Calibration Environment," *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 10, pp. 9103–9110, 2024.
- [4] R. Marcuzzi, L. Nunes, L. Wiesmann, J. Behley, and C. Stachniss, "Mask-Based Panoptic LiDAR Segmentation for Autonomous Driving," *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 2, pp. 1141–1148, 2023.
- [5] R. Marcuzzi, L. Nunes, L. Wiesmann, E. Marks, J. Behley, and C. Stachniss, "Mask4D: End-to-End Mask-Based 4D Panoptic Segmentation for LiDAR Sequences," *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 11, pp. 7487–7494, 2023.
- [6] L. Wiesmann, L. Nunes, J. Behley, and C. Stachniss, "KPPR: Exploiting Momentum Contrast for Point Cloud-Based Place Recognition," *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 2, pp. 592–599, 2023.
- [7] X. Chen, B. Mersch, L. Nunes, R. Marcuzzi, I. Vizzo, J. Behley, and C. Stachniss, "Automatic Labeling to Generate Training Data for Online LiDAR-Based Moving Object Segmentation," *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, pp. 6107–6114, 2022.
- [8] R. Marcuzzi, L. Nunes, L. Wiesmann, I. Vizzo, J. Behley, and C. Stachniss, "Contrastive Instance Association for 4D Panoptic Segmentation using Sequences of 3D LiDAR Scans," *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 1550–1557, 2022.
- [9] B. Mersch, X. Chen, I. Vizzo, L. Nunes, J. Behley, and C. Stachniss, "Receding Moving Object Segmentation in 3D LiDAR Data Using Sparse 4D Convolutions," *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, pp. 7503–7510, 2022.
- [10] L. Nunes, R. Marcuzzi, X. Chen, J. Behley, and C. Stachniss, "SegContrast: 3D Point Cloud Feature Representation Learning through Self-supervised Segment Discrimination," *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 2116–2123, 2022.
- [11] L. Nunes, X. Chen, R. Marcuzzi, A. Osep, L. Leal-Taixé, C. Stachniss, and J. Behley, "Unsupervised Class-Agnostic Instance Segmentation of 3D LiDAR Data for Autonomous Vehicles," *IEEE Robotics and Automation Letters (RA-L)*, 2022.

Peer-reviewed conference papers

- [1] R. Marcuzzi, L. Nunes, E. Marks, X. Zhong, J. Behley, and C. Stachniss, "Vision-Based Panoptic Occupancy Prediction in Urban Environments," in *Proceedings of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2026.
- [2] Y. Chong, L. Nunes, F. Magistri, X. Zhong, J. Behley, and C. Stachniss, "Zero-Shot Semantic Segmentation for Robots in Agriculture," in *Proceedings of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2025.
- [3] E. Marks, L. Nunes, F. Magistri, M. Sodano, R. Marcuzzi, L. Zimmermann, J. Behley, and C. Stachniss, "Tree Skeletonization from 3D Point Clouds by Denoising Diffusion," in *Proceedings of the IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2025.
- [4] L. Nunes, R. Marcuzzi, B. Mersch, J. Behley, and C. Stachniss, "Scaling Diffusion Models to Real-World 3D LiDAR Scene Completion," in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [5] M. Sodano, F. Magistri, L. Nunes, J. Behley, and C. Stachniss, "Open-World Semantic Segmentation Including Class Similarity," in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [6] H. Lim, L. Nunes, B. Mersch, X. Chen, J. Behley, H. Myung, and C. Stachniss, "ERASOR2: Instance-Aware Robust 3D Mapping of the Static World in Dynamic Scenes," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [7] L. Nunes, L. Wiesmann, R. Marcuzzi, X. Chen, J. Behley, and C. Stachniss, "Temporal Consistent 3D LiDAR Representation Learning for Semantic Perception in Autonomous Driving," in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [8] I. Vizzo, B. Mersch, L. Nunes, L. Wiesmann, T. Guadagnino, and C. Stachniss, "Toward Reproducible Version-Controlled Perception Platforms: Embracing Simplicity in Autonomous Vehicle Dataset Acquisition," in *Proc. of the Int. Conf. on Intelligent Transportation Systems Workshops*, accepted, 2023.
- [9] D. Bruno, L. P. N. Matias, J. Amaro, F. S. Osório, and D. Wolf, "Computer Vision System with 2D and 3D Data Fusion for Detection of Possible Auxiliaries Routes in Stretches of Interdicted Roads," in *Proceedings of the Annual Hawaii Int. Conf. on System Sciences (HICSS)*, 2019, pp. 7372–7381.
- [10] L. P. N. Matias, M. Sons, J. R. Souza, D. F. Wolf, and C. Stiller, "VeIGAN: Vectorial Inpainting Generative Adversarial Network for Depth Maps Object Removal," in *2019 IEEE Intelligent Vehicles Symposium (IV)*, 2019, pp. 310–316.
- [11] L. P. N. Matias, T. C. Santos, D. F. Wolf, and J. R. Souza, "Trajectory Planning for UGV Using Clothoids," in *Robotics*, F. Santos Osório and R. Sales Gonçalves, Eds., Springer International Publishing, 2016, pp. 281–298.
- [12] L. P. N. Matias, T. C. Santos, D. F. Wolf, and J. R. Souza, "Path Planning and Autonomous Navigation using AMCL and AD," in *2015 12th Latin American Robotics Symposium and 2015 3rd Brazilian Symposium on Robotics (LARS-SBR)*, 2015, pp. 320–324.

Preprint papers

- [1] T. Faasch, J. Kall, L. Nunes, J. Behley, and C. Stachniss, "TASE: Truncation-Aware Semantic Embeddings for 3D Scene Understanding and Editing," *arXiv Preprint*, vol. arxiv:2606.03314, 2026.
- [2] K. Knaebel, G. M. Garcia, C. Schmidt, I. Fradlin, L. Nunes, D. de Geus, and B. Leibe, "SurGe: Improved Surface Geometry in Point Maps," *arXiv Preprint*, vol. arxiv:2605.31577, 2026.
- [3] Y. Pan, T. Sun, L. Zhu, L. Nunes, I. Armeni, J. Behley, and C. Stachniss, "Register Any Point: Scaling 3D Point Cloud Registration by Flow Matching," *arXiv Preprint*, vol. arXiv:2512.01850, 2026.
- [4] L. P. N. Matias, J. R. Souza, and D. F. Wolf, "Environment reconstruction on depth images using Generative Adversarial Networks," *arXiv Preprint*, vol. arxiv:1912.03992, 2019.

Theses

- [1] L. Nunes, "Learning Discriminative Representations and Generative Approaches for Outdoor 3D LiDAR Data," PhD thesis, University of Bonn, 2026.
- [2] L. Nunes, "Environment reconstruction on disparity images using surface features and Generative Adversarial Networks," Master's thesis, University of São Paulo, 2020.